# Additional Assignment: Real-time Stock Data (Kafka / SSE)

> **Estimated time:** ~60 minutes
> **Tools allowed:** any AI coding assistant (Cursor AI, VS Code + Copilot, etc.) — vibe coding encouraged!

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## 1. Context

During this lab you will consume a **live stream of fictitious stock price ticks** for 50 companies.
The data is produced by a Kafka broker running on the instructor's server and exposed via a simple **HTTP / SSE API** — no Kafka client libraries required.

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## 2. Step 1 — Get your API key

Open in the browser:

```
https://add.piotrkojalowicz.dev/
```

1. Enter the **shared class password**: `A@d-$01`
2. Copy the generated **API key** — it is **unique per student**.
   You will need it in every API call.

> ⚠️ The service is open only during class hours (until 19:00).
> Rate limit: first 10 requests free, then max 1 request / 10 s per key.

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## 3. Available API endpoints

Base URL: `https://add.piotrkojalowicz.dev`

## Authentication

Every request must include the key as a header **or** a query param:

| Method | Example |
|---|---|
| Header | `X-API-Key: <YOUR_KEY>` |
| Query param | `?api_key=<YOUR_KEY>` (convenient for browser) |

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**`GET /api/tickers`**

Returns the full list of 50 companies.

```
curl -H "X-API-Key: <KEY>" "https://add.piotrkojalowicz.dev/api/tickers"
```

Browser:

```
https://add.piotrkojalowicz.dev/api/tickers?api_key=<KEY>
```

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**`GET /api/latest?ticker=ACME`**

Returns the most recent tick for one or more tickers.

```
curl -H "X-API-Key: <KEY>" \
  "https://add.piotrkojalowicz.dev/api/latest?ticker=ACME&ticker=ALFA"
```

Browser:

```
https://add.piotrkojalowicz.dev/api/latest?ticker=ACME&ticker=ALFA&api_key=<KEY>
```

### Response format:

```
{
  "data": [
    {
      "ticker": "ACME",
```

```
      "ts": "2026-05-07T15:30:00+02:00",
      "price": 123.45,
      "currency": "PLN",
      "volume": 1200,
      "seq": 42
    }
  ]
}
```

> Data is retained for **~10 minutes** only — older ticks are automatically deleted.

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`GET /api/stream?ticker=ACME`

Live stream via **Server-Sent Events (SSE)** — new tick sent every ~second.

```
curl -N -H "X-API-Key: <KEY>" \
  "https://add.piotrkojalowicz.dev/api/stream?ticker=ACME"
```

Browser (live view):

```
https://add.piotrkojalowicz.dev/api/stream?ticker=ACME&api_key=<KEY>
```

Events arrive as:

```
event: tick
data: {"ticker":"ACME","ts":"...","price":124.10,...}
```

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# 4. Task — Build 2 small apps

Use any language and framework. AI coding tools are recommended.

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### App #1: Realtime Dashboard

**Goal:** show live price updates for selected companies.

Requirements: - User can **select one or more tickers** from the list - App connects to `/api/stream` and shows **live updates** in the UI - Each update shows at minimum: ticker, price, timestamp - Optionally: short price history (last 20–30 ticks) or a mini chart

**Hint:** Use `EventSource` (JavaScript) or an SSE client library in your language.

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### App #2: History Downloader / Viewer

**Goal:** fetch recent data, display it, and save it locally.

Requirements: - User can **select tickers** and a **time range** (e.g. last 1–10 minutes) - App fetches data from `/api/latest` (repeatedly, with a delay ≥10 s) or `/api/stream` - Displays results as a **table or chart** in the UI - Allows **saving to CSV or JSON**

> Remember: data older than ~10 minutes is gone. Design your polling accordingly.

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## 5. Deliverables

Submit (e.g. as a GitHub repo or ZIP):

- [ ] Source code for both apps
- [ ] `README.md` describing:
- how to run locally
- how the API key is configured (env variable, `.env` file, etc.)
- which endpoints are used and how
- [ ] The API key must **not** be hardcoded in the repository

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## 6. Grading

See `ACCEPTANCE.md` for the full checklist.

| Criterion | Weight |
|---|---|
| App #1 works (live stream, ticker selection, UI) | 40% |
| App #2 works (data fetch, time range, save to file) | 40% |

| Criterion | Weight |
|---|---|
| Code quality & README | 20% |

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## 7. List of tickers

| Ticker | Company |
|---|---|
| ACME | ACME Corp. |
| ALFA | Alfa Technologies |
| BETA | Beta Retail Group |
| CASH | CashBank |
| CLOUD | CloudNine |
| COAL | Coal Energy |
| COPR | Copper Mining Co. |
| DATA | DataWorks |
| DEVS | DevStudio |
| DRON | Dronix |
| ECO | EcoPower |
| EDU | EduNext |
| ENRG | Energo |
| FARM | FarmFoods |
| FINX | FinX |
| FOOD | FoodBox |
| FUEL | FuelOne |
| GAME | GameForge |
| GRIN | Green Invest |

| Ticker | Company |
|---|---|
| HEAL | HealTech |
| HOME | HomeBuild |
| HYPE | Hype Media |
| INSR | InsureCo |
| IOT | IoT Systems |
| JET | Jet Logistics |
| LABS | Labs Research |
| LEND | Lendify |
| LOGI | LogiWare |
| MALL | Mall Retail |
| MEDI | MediCare |
| META | MetaCom |
| MOBI | MobiTel |
| MOVE | MoveNow |
| NET | Netlink |
| NOVA | Nova Ventures |
| OILS | OilSands |
| PARK | Park Realty |
| PHAR | Pharmax |
| PLNT | Plantio |
| PROD | Prodigo |
| QBIT | QBit Quantum |
| RAIL | Rail Cargo |
| ROBO | RoboMakers |

| Ticker | Company |
|---|---|
| SAFE | SafeSecurity |
| SHIP | ShipIt |
| SHOP | ShopNow |
| SOLR | Solaris |
| TEL | TelcoPlus |
| TRVL | TravelBee |
| WATR | WaterWorks |

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# 8. Quick-start example (Python)

```
import requests, os

API_KEY = os.environ["ADD_API_KEY"]   # set in .env or shell
BASE    = "https://add.piotrkojalowicz.dev"
HEADERS = {"X-API-Key": API_KEY}

# list tickers
tickers = requests.get(f"{BASE}/api/tickers", headers=HEADERS).json()
print(tickers["tickers"][:5])

# latest price for ACME
data = requests.get(f"{BASE}/api/latest", params={"ticker": "ACME"},
headers=HEADERS).json()
print(data)
```

SSE stream (Python):

```
import sseclient, requests, os

API_KEY = os.environ["ADD_API_KEY"]
resp = requests.get(
    "https://add.piotrkojalowicz.dev/api/stream",
    params={"ticker": "ACME"},
    headers={"X-API-Key": API_KEY},
    stream=True,
```

```
)
for event in sseclient.SSEClient(resp):
    if event.event == "tick":
        print(event.data)
```

JavaScript (browser / Node):

```
const key = "YOUR_KEY";
const es  = new EventSource(
  `https://add.piotrkojalowicz.dev/api/stream?ticker=ACME&api_key=${key}`
);
es.addEventListener("tick", e => console.log(JSON.parse(e.data)));
```